create the opportunity for more efficient and effective market operations,[3] and that it is in the public interest and appropriate for the protection of investors and the maintenance of fair and orderly markets to ensure the availability to brokers, dealers, and investors of information with respect to quotations for and transactions in securities.[4] Section 11A of the Exchange Act also authorizes the Commission to prescribe rules to ensure the "prompt, accurate, reliable, and fair collection, processing, distribution, and publication of information with respect to quotations for and transactions in such securities and the fairness and usefulness of the form and content of such information."[5] In furtherance of these purposes, the Commission has sought through its rules and regulations to help ensure that certain "core data"[6] is widely available for reasonable fees.[7] The Commission has recognized that investors must have certain core data "to participate in the U.S. equity markets."[8]

On February 14, 2020, the Commission proposed to amend Regulation NMS to better achieve the goal of Section 11A of the Exchange Act of assuring "the availability to brokers, dealers, and investors of information with respect to quotations for and transactions in securities"

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3 See 15 U.S.C. 78k-1(a)(1)(B). See also S. Rep. No. 94-75, 94th Cong., 1st Sess. (1975) (noting that the systems for collecting and distributing consolidated market data would "form the heart of the national market system").

4 See 15 U.S.C. 78k-1(a)(1)(C).

5 15 U.S.C. 78k-1(c)(1)(B).

6 See infra note 17 and accompanying text (defining "core data").

7 See Rule 603 of Regulation NMS; see also, e.g., Securities Exchange Act Release No. 51808 (June 9, 2005), 70 FR 37496, 37560 (June 29, 2005) ("Regulation NMS Adopting Release") ("In the Proposing Release, the Commission emphasized that one of its primary goals with respect to market data is to assure reasonable fees that promote the wide public availability of consolidated market data.").

8 Regulation NMS Adopting Release, supra note 7, at 37560.

that is prompt, accurate, reliable, and fair.[9] The amendments as adopted endeavor to fulfill this goal of Section 11A of the Exchange Act by updating the content of "core data" and the manner in which it is provided to investors in the national market system.

A. Current Market Data Content and Dissemination Model Under Regulation NMS

The Commission established many of the current requirements of the national market system under Regulation NMS and approved the three effective national market system plans shortly after Congress enacted Section 11A in the 1975 amendments to the Exchange Act ("1975 Amendments").[10] Under Regulation NMS and the Equity Data Plans,[11] the self-regulatory

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9 See Securities Exchange Act Release No. 88216 (Feb. 14, 2020), 85 FR 16726, 27 (Mar. 24, 2020) ("Market Data Infrastructure Proposing Release" or "Proposing Release").

10 The three effective national market system plans that govern the collection, consolidation, processing, and dissemination of certain NMS information are: (1) the Consolidated Tape Association Plan ("CTA Plan"); (2) the Consolidated Quotation Plan ("CQ Plan"); and (3) the Joint Self-Regulatory Organization Plan Governing the Collection, Consolidation, and Dissemination of Quotation and Transaction Information for Nasdaq-Listed Securities Traded on Exchanges on an Unlisted Trading Privileges Basis ("UTP Plan") (together, the "Equity Data Plans"). Each of the Equity Data Plans is an effective national market system plan under 17 CFR 242.608 (Rule 608) of Regulation NMS. See also Securities Exchange Act Release Nos. 10787 (May 10, 1974), 39 FR 17799 (order approving CTA Plan); 15009 (July 28, 1978), 43 FR 34851 (Aug. 7, 1978) (order temporarily approving CQ Plan); 16518 (Jan. 22, 1980), 45 FR 6521 (Jan. 28, 1980) (order permanently approving CQ Plan); 28146 (June 26, 1990), 55 FR 27917 (July 6, 1990) (order approving UTP Plan). The options exchanges are participants in the Limited Liability Company Agreement of Options Price Reporting Authority, LLC ("OPRA Plan"), a plan under Rule 608 of Regulation NMS, which governs the collection, consolidation, processing, and dissemination of last sale and quotation information for listed options. See Securities Exchange Act Release Nos. 17638 (Mar. 18, 1981), 22 S.E.C. Docket 484 (Mar. 31, 1981); 61367 (Jan. 15, 2010), 75 FR 3765 (Jan. 22, 2010).

11 Rule 603(b) of Regulation NMS, 17 CFR 242.603(b), requires that every national securities exchange on which an NMS stock is traded and national securities association act jointly pursuant to one or more effective national market system plans to disseminate consolidated information on quotations for and transactions in NMS stocks, and that such

organizations ("SROs") are required to provide certain quotation[12] and transaction information[13] for each NMS stock to an exclusive plan processor ("exclusive SIP"),[14] which consolidates this information and makes it available to market participants on the consolidated tapes.[15] For each NMS stock, the Equity Data Plans currently provide for the dissemination of top-of-book ("TOB") data and transaction information, generally defining consolidated market information (or "core data") as consisting of: (1) the price, size, and exchange of the last sale; (2) each exchange's current highest bid and lowest offer and the shares available at those prices; and (3) the national best bid and national best offer ("NBBO")[16] (i.e., the highest bid and lowest offer currently available on any exchange).[17] In addition to disseminating core data, the exclusive

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plan or plans provide for the dissemination of all consolidated information for an individual NMS stock through a single plan processor.

12 See Rule 602 of Regulation NMS.

13 See Rule 601 of Regulation NMS, 17 CFR 242.601.

14 See Rule 600(b)(67) of Regulation NMS, 17 CFR 242.600(b)(67) (defining plan processor). See also Section 3(a)(22)(B) of the Exchange Act, 15 U.S.C. 78c(22)(B) (defining exclusive processor).

15 The Equity Data Plans disseminate SIP data over three separate networks: (1) Tape A for securities listed on the New York Stock Exchange ("NYSE"); (2) Tape B for securities listed on exchanges other than NYSE and Nasdaq; and (3) Tape C for securities listed on Nasdaq. These tapes are referred to as the "consolidated tapes." The CTA Plan governs the collection, consolidation, processing, and dissemination of last sale information for Tape A and Tape B securities. The CQ Plan governs the collection, consolidation, processing, and dissemination of quotation information for Tape A and Tape B securities. Finally, the UTP Plan governs the collection, consolidation, processing, and dissemination of last sale and quotation information for Tape C securities.

16 See Rule 600(b)(50) of Regulation NMS for the definition of NBBO.

17 See Bloomberg Order, infra note 22, at 3; see also Rescission of Effective-Upon-Filing Procedures for NMS Plan Fee Amendments, Securities Exchange Act Release No. 89618 (Aug. 19, 2020), 85 FR 65470 (Oct. 15, 2020) ("Effective-Upon-Filing Adopting Release").

SIPs collect, calculate, and disseminate certain regulatory data—including information required by the National Market System Plan to Address Extraordinary Market Volatility ("LULD Plan"),[18] information relating to regulatory halts and market-wide circuit breakers, and information regarding the short-sale price test pursuant to Rule 201 of Regulation SHO.[19] They also collect and disseminate other NMS information and disseminate certain administrative messages.[20] Together with core data, the Commission refers to this broader set of data for purposes of this release as "SIP data."

The purpose of the Equity Data Plans, approved under Regulation NMS, is to facilitate the collection and dissemination of SIP data so that the public has ready access to a "comprehensive, accurate, and reliable source of information for the prices and volume of any NMS stock at any time during the trading day."[21] Widespread availability of timely market information promotes fair and efficient markets and facilitates the ability of brokers and dealers to provide best execution to their customers.[22] Many of the requirements under Regulation NMS

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18 See Limit Up Limit Down Plan, available at http://www.luldplan.com (last accessed Sept. 24, 2020).

19 Rule 201(b)(3).

20 For example, messages regarding cancelled and erroneous trades are included in the data disseminated by the exclusive SIPs. See, e.g., Consolidated Tape System, Multicast Output Binary Specification, 36, 47 (October 2, 2020), available at https://www.ctaplan.com/publicdocs/ctaplan/CTS_Pillar_Output_Specification.pdf.

21 Concept Release on Equity Market Structure, Securities Exchange Act Release No. 61358 (Jan. 14, 2010), 75 FR 3593 (Jan. 21, 2010).

22 See In the Matter of the Application of Bloomberg L.P., Securities Exchange Act Release No. 83755 at 3 (July 31, 2018), available at https://www.sec.gov/litigation/opinions/2018/34-83755.pdf ("Bloomberg Order"); SEC Concept Release: Regulation of Market Information Fees and Revenues, Securities Exchange Act Release No. 42208 (Dec. 9, 1999), 64 FR 70613, 70615 (Dec. 17, 1999) ("Market Information Concept Release") (stating that the distribution of core data "is the

and the Equity Data Plans that establish the national market system have not been updated since their adoption despite dramatic changes in the operation of the market and market participants' information needs.[23]

In addition to the SIP data provided via the Equity Data Plans, most exchanges have developed many proprietary TOB products that contain the quotation and transaction data that they provide to the exclusive SIPs as well as proprietary depth-of-book ("DOB") products that contain more extensive information that is not provided by the exclusive SIPs, such as complete order-by-order information, full depth of book information, auction information, and odd-lot quotation information.[24] The exchanges provide individual exchange proprietary data products directly to market participants and sometimes consolidate them with their affiliated exchanges' proprietary data feeds. The exchanges make these proprietary data products available with different connectivity and transmission options, many of which are faster than those available for the consolidated tapes. Market participants that purchase proprietary DOB data feeds directly generally aggregate the information in a decentralized manner in an effort to create a

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principal tool for enhancing the transparency of the buying and selling interest in a security, for addressing the fragmentation of buying and selling interest among different market centers, and for facilitating the best execution of customers' orders by their broker-dealers").

23 See Proposing Release, 85 FR at 16728, n. 13 and accompanying text.

24 While the pre-Regulation NMS rules permitted the independent distribution of quotes by individual SROs, Rule 603(a) of Regulation NMS, 17 CFR 242.603(a), was adopted to impose "uniform standards" on such distribution (i.e., the "fair and reasonable" and "not unreasonably discriminatory" standards). See Regulation NMS Adopting Release, supra note 7, at 37569. Prior to Regulation NMS, however, SROs and their members were prohibited from disseminating their trade reports independently. Id. at 37589.

consolidated view of the market that is both more timely and more complete than the exclusive SIP data feeds provided by the Equity Data Plans.

As discussed further below, Regulation NMS and the Equity Data Plans have not kept pace with the business demands of market participants.[25] While the exchanges have developed individual proprietary data products to meet the needs of some market participants, the Commission believes that there should be improvement to, and modernization of, the national market system to fulfill the goals of Section 11A of the Exchange Act and to meet the current core data demands of market participants.[26] Over the last 15 years, the exchanges have moved

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25 See infra Section III.A.

26 Commenters generally expressed concern that SIP data provided by the Equity Data Plans was not sufficient for some market participants. See, e.g., letters to Vanessa Countryman, Secretary, Commission, from Mehmet Kinak, Vice President and Global Head of Systematic Trading and Market Structure, and Jonathan D. Siegel, Vice President and Senior Legal Counsel, Legislative and Regulatory Affairs, T. Rowe Price, dated June 3, 2020, ("T. Rowe Price Letter") at 1 ("Unfortunately, as the SIPs have not kept pace with the dramatic technological and market developments over the past decade, they are no longer satisfying the needs of a broad cross-section of market participants. Due to its limited content and higher latency, the usage of SIP data is adequate only for investors that visually consume NMS information (e.g., humans looking at quotes on a screen"); Thomas M. Merritt, Deputy General Counsel, Virtu Financial, Inc., dated May 26, 2020, ("Virtu Letter") at 2 ("the 'core data' offered through the SIPs is no longer sufficient for most market participants to trade competitively in today's market place."), 5; Michael Blasi, Vice President, Enterprise Infrastructure, and Krista Ryan, Vice President and Associate General Counsel, Fidelity Investments, dated May 26, 2020, ("Fidelity Letter") at 2 ("the SIPs have not kept pace with the U.S. equity markets which, through technological and market developments, now offer more products, faster, and at a lower cost."); Joseph J. Barry, Senior Vice President and Global Head of Regulatory, Industry, and Government Affairs, State Street Corporation, dated May 26, 2020, ("State Street Letter") at 2 (". . . regulatory obligations and customer expectations related to best execution, transaction cost analysis, transparency and market competition generated further need for data that is unavailable on the SIPs. As a result, market participants have become increasingly dependent on proprietary data feeds marketed by the exchanges outside of the SIPs."); Hubert De Jesus, Managing Director, Global Head of Market Structure and Electronic Trading, and Samantha DeZur, Director, Global Public Policy,

from largely manual, floor-based models to predominantly electronic trading systems and market participants have likewise largely incorporated sophisticated, latency-sensitive, and data dependent electronic trading technologies for their trading needs. This has contributed to some market participants stating that they require additional, and more timely, information for their best execution analysis.[27] The Commission agrees that more comprehensive and latency-

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BlackRock, Inc., dated May 26, 2020, ("BlackRock Letter") at 1 ("However, the current model for and content of NMS market data has not kept pace with the evolution in equity markets and correspondingly the quality of the Securities Information Processors ("SIPs") has declined, lowering public confidence in the market."); Jennifer W. Han, Associate General Counsel, Managed Funds Association, dated May 29, 2020, ("MFA Letter") at 2 ("Today, the current exclusive SIP model and content of core data does not serve the needs of investors, many of whom must subscribe to the exchanges' proprietary market data feeds at considerable additional cost to trade effectively, while others are forced to rely on inferior information and outdated technology."); Peter D. Stutsman, Global Equity Trading Manager, The Capital Group Companies, Inc., dated June 2, 2020, ("Capital Group Letter") at 2 ("Over the last 15 years, the discrepancy in data elements and latency between proprietary feeds and the consolidated tape has expanded such that the SIP is no longer a realistic tool for institutional investors or broker-dealers in meeting their respective best execution obligations when routing orders."); Makan Delrahim, Assistant Attorney General, U.S. Department of Justice, Antitrust Division, Rene L. Augustine, Deputy Assistant Attorney General, Michael F. Murray, Deputy Assistant Attorney General, David B. Lawrence, Chief, Karina B. Lubell, Assistant Chief, Charles J. Ramsey, Attorney, Antitrust Division Competition Policy and Advocacy Section, and Ihan Kim, Attorney, Technology and Financial Services Section, dated May 26, 2020, ("DOJ Letter"); Mark Garabedian, Manager, Trading Data and Analytics, and Lisa Mahon Lynch, Associate Director, Global Trading, Wellington Management Company LLP, dated May 27, 2020, ("Wellington Letter").

[27] See, e.g., letters to Vanessa Countryman, Secretary, Commission, from Lev Bagramian, Senior Securities Policy Advisor, Better Markets, Inc., dated May 26, 2020, ("Better Markets Letter") at 1–2; Joe Wald and Ray Ross, Managing Directors, BMO Capital Markets Group and Co-Heads of Electronic Trading, Clearpool, dated June 2, 2020, ("Clearpool Letter") at 1, 11; John Ramsay, Chief Market Policy Officer, Investors Exchange LLC, dated May 28, 2020, ("IEX Letter") at 5; Jim Considine, Chief Financial Officer, McKay Brothers LLC, dated May 31, 2020, ("McKay Letter") at 1; Rich Steiner, Head of Client Advocacy and Market Innovation, RBC Capital Markets, LLC, dated May 27, 2020, ("RBC Letter") at 4; Ellen Greene, Managing Director, Equity and Options Market Structure, SIFMA, dated May 26, 2020, ("SIFMA Letter") at 3–4; Capital Group

sensitive NMS information can be significantly beneficial in facilitating informed trading decisions, and the Commission believes that such information should be more widely distributed and more readily accessible. Further, while the proprietary DOB products provided by exchanges contain the data elements included within expanded core data, commenters have stated that the cost of these proprietary market data products inhibits the purchase of, and the widespread dissemination of, this data to market participants that may need it to participate effectively in the markets.[28] The Commission is concerned that the two different methods of

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Letter at 2; DOJ Letter at 2, 4; State Street Letter at 2; T. Rowe Price Letter at 1; Virtu Letter at 2.

28 See, e.g., Virtu Letter at 5 ("[I]ncluding depth of book information in the SIP will allow investors who cannot afford to pay for costly Exchange proprietary feeds to trade more competitively in the marketplace . . . ."); SIFMA Letter at 2 ("[W]e do not believe that the SIPs currently provide the necessary data to market participants at the requisite speed to efficiently trade in today's high speed and automated marketplace. As a result, many broker-dealers, asset managers and other market participants are forced to purchase proprietary data feeds from individual exchanges to create a consolidated and robust view of the market, while additionally bearing the economic burden of having to purchase consolidated data from the SIPs. This results in an enormous cost burden on the marketplace and creates a two-tiered market for market data by limiting access to critical market data at the fastest speeds to those who can afford to pay the exorbitant fees charged for it by the exchanges."); MFA Letter at 2 ("Today, the current exclusive SIP model and content of core data does not serve the needs of investors, many of whom must subscribe to the exchanges' proprietary market data feeds at considerable additional cost to trade effectively, while others are forced to rely on inferior information and outdated technology."); Clearpool Letter at 2 ("As we have stated on a number of previous occasions, of all the issues relating to the costs of trading, the trend toward higher market data fees has had the most negative impact on the securities markets. It remains increasingly difficult for many broker-dealers to compete in the current market environment due, in part, to issues related to the costs associated with trading."); Dorothy Donohue, Deputy General Counsel, Securities Regulation, Investment Company Institute, dated May 26, 2020, ("ICI Letter") at 9–10 ("Including auction information in the consolidated feed would enhance transparency into market activity. Doing so also would eliminate proprietary data costs as a barrier to auction trading and encourage a broader range of market participants to submit trading interest.").

data dissemination—SIP data provided pursuant to Regulation NMS and the Equity Data Plans and proprietary data products provided by the exchanges—have contributed to the development of a two-tiered data market that raises fundamental concerns about the ability of the national market system to continue to ensure that the goals of Section 11A of the Exchange Act are being met, including: (i) fair competition among brokers and dealers;[29] (ii) the availability to brokers, dealers, and investors of NMS information;[30] and (iii) the practicability of brokers executing investors' orders at the best available prices.[31] Section 11A of the Exchange Act directs the Commission to facilitate the establishment of a national market system in accordance with these, and other, Congressional findings. Therefore, the Commission believes that Regulation NMS should be amended to update the national market system in accordance with the findings and to carry out the objectives set forth in Section 11A and to "assure the prompt, accurate, reliable, and fair collection, processing, distribution, and publication" of NMS information and "the fairness and usefulness of the form and content of such information."[32]

B. National Market System Initiatives and the Market Data Infrastructure Proposing Release

The Commission has monitored the national market system and its operation in light of changes in the markets and, over the years, has observed increased concerns about the usefulness, fairness, and promptness of the consolidated tapes. The Division of Trading and

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29 Section 11A(a)(1)(C)(ii) of the Exchange Act, 15 U.S.C. 78k-1(a)(1)(C)(ii).

30 Section 11A(a)(1)(C)(iii) of the Exchange Act, 15 U.S.C. 78k-1(a)(1)(C)(iii).

31 Section 11A(a)(1)(C)(iv) of the Exchange Act, 15 U.S.C. 78k-1(a)(1)(C)(iv).

32 Section 11A(c)(1)(B) of the Exchange Act, 15 U.S.C. 78k-1(c)(1)(B).

Markets held a Roundtable on Market Data in October of 2018,[33] at which some market participants discussed their views about the shortcomings of the existing centralized consolidation model and the need for updates to the national market system to reflect the now widespread use of electronic trading and the need for more, faster NMS information.[34]

Further, the Commission has considered how the provision of the current consolidated tapes and proprietary data feeds has affected investors' access to NMS information. The Commission understands that different types of investors have different information needs. However, as stated above, the Commission is concerned that a two-tiered system has developed in which certain market participants who are able to afford, and choose to pay for, the exchanges' proprietary DOB data feeds and associated connectivity and transmission offerings receive more content-rich data faster than those who do not receive these data feeds, such as market participants that face higher barriers to entry from data and other exchange fees.[35] Market participants that do not receive proprietary DOB feeds may be affected in their efforts to seek best execution and otherwise effectively compete with market participants that receive proprietary DOB data feeds because they do not obtain access to the additional content and may be receiving data in a slower manner.

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33 See Equity Market Structure Roundtables, Oct. 25–26, 2018: Roundtable on Market Data and Market Access, SEC, available at https://www.sec.gov/spotlight/equity-market-structure-roundtables ("Market Data Roundtable").

34 See Proposing Release, 85 FR at 16765, n. 393 and accompanying text.

35 See Proposing Release, 85 FR at 16768. See also infra Section V.B.3(b). Proprietary data fees have increased over the last decade, and are generally more expensive relative to SIP data fees, and there are indicia that exchanges may not be subject to robust competition with respect to market data. See infra notes 1780-1788 and accompanying text.